

計算：

2025 Home Work

Ex. 4.7 For a balanced three-phase transmission line with $z=1\angle\theta_z, \theta_z=10^\circ$,

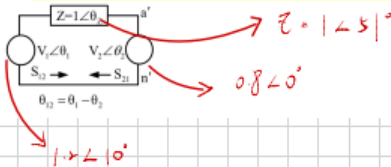
let $|V_1|=1.2$ and $|V_2|=0.8$

(a) Plot S_{12} and $-S_{21}$

(b) Find P_{12max} and $-P_{21max}$

θ_z : 學號末二位數在50-99, θ_z = 學號末二位數, 例如學號98310073則 $\theta_z = 73^\circ$

學號末二位數在00-49, θ_z = 學號末二位數+50, 例如學號98310035則 $\theta_z = 35 + 50 = 85^\circ$



$$S_{12} = \frac{|V_1|^2}{|Z|} \angle \theta_z - \frac{|V_1||V_2|}{|Z|} \angle \theta_z \angle \theta_{12}$$

$$= \frac{1.2^2}{1} \angle 51^\circ - \frac{1.2 \times 0.8}{1} \angle 61^\circ = \underline{0.4408} + j \underline{0.2795}$$

P_{12}

$$-S_{21} = \frac{|V_2|^2}{|Z|} \angle \theta_z + \frac{|V_1||V_2|}{|Z|} \angle \theta_z \angle -\theta_{12}$$

$$= \frac{0.8^2}{1} \angle 51^\circ + \frac{0.96}{1} \angle 41^\circ = \underline{0.3218} + j \underline{0.324}$$

$-P_{21}$

if $\theta_{12} = 0^\circ$

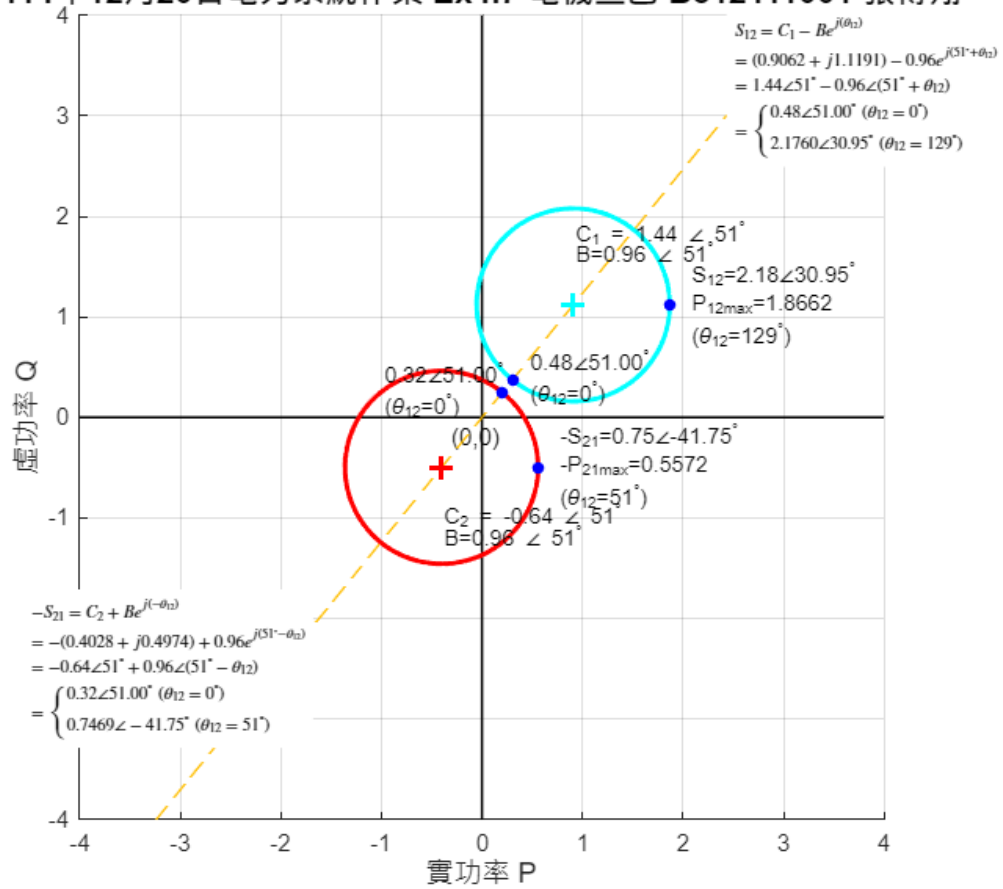
$$\rightarrow S_{12} = 0.3201 + j 0.3730 = 0.48 \angle 51^\circ$$

$$-S_{21} = 0.2014 + j 0.2487 = 0.32 \angle 51^\circ$$

B812111001 張博翔

短程輸電線複數功率傳輸之圓線圖：

114年12月20日電力系統作業 Ex4.7 電機三乙 B812111001 張博翔



程式碼：

```
clc; clear; close all;

student_id = 01;
full_id = 'B812111001';
my_name = '張博翔';
class_name = '電機三乙';

% 計算
theta_z_deg = student_id + 50;
theta_z_rad = deg2rad(theta_z_deg);

V1 = 1.2;
V2 = 0.8;
abs_Z = 1.0;

% 圓心與半徑
% C1 (送電端圓心)  $V1^2 / Z^*$ 
C1 = (V1^2) / (abs_Z * exp(-1j * theta_z_rad));
C1_mag = abs(C1);

% C2 (受電端圓心)  $-V2^2 / Z^*$ 
C2 = - (V2^2) / (abs_Z * exp(-1j * theta_z_rad));
C2_mag = abs(C2);

% |B| (半徑)  $V1*V2/Z$ 
B_val = (V1 * V2) / abs_Z;

% 圓軌跡
delta = linspace(0, 2*pi, 360);

% S12 = C1 - |B| * e^j(theta_z + theta12)
```

```

S12_circle = C1 - B_val * exp(1j * (theta_z_rad + delta));
% -S21 = C2 + |B| * e^j(theta_z - theta12)
nS21_circle = C2 + B_val * exp(1j * (theta_z_rad - delta));

% theta12 = 0
S12_zero = C1 - B_val * exp(1j * (theta_z_rad + 0));
nS21_zero = C2 + B_val * exp(1j * (theta_z_rad - 0));

% P12 max
theta_P12max = 180 - theta_z_deg;
P12_max_val = real(C1) + B_val;
S12_max_point = C1 - B_val * exp(1j * deg2rad(theta_z_deg +
theta_P12max));

% -P21 max
theta_nP21max = theta_z_deg;
nP21_max_val = real(C2) + B_val;
nS21_max_point = C2 + B_val * exp(1j * deg2rad(theta_z_deg -
theta_nP21max));

figure('Color', 'w', 'Position', [100, 50, 900, 800]);

hold on; grid on; axis equal;
axis([-4 4 -4 4]);
xlabel('實功率 P', 'FontSize', 12);
ylabel('虛功率 Q', 'FontSize', 12);

title_str = sprintf('114 年 12 月 20 日電力系統作業 Ex4.7 %s %s %s',
class_name, full_id, my_name);
title(title_str, 'FontSize', 14, 'FontWeight', 'bold');

plot([-5 5], [0 0], 'k', 'LineWidth', 1); % X 軸
plot([0 0], [-5 5], 'k', 'LineWidth', 1); % Y 軸
text(-0.3, -0.2, '(0,0)', 'FontSize', 10);

```

```
% 阻抗角線 theta_z (51 度)
```

```
L = 6;
```

```
plot([-L*cos(theta_z_rad), L*cos(theta_z_rad)], ...  
      [-L*sin(theta_z_rad), L*sin(theta_z_rad)], ...  
      'Color', [1, 0.75, 0], 'LineStyle', '--', 'LineWidth', 0.5);
```

```
% 繪製圓與圓心
```

```
plot(real(S12_circle), imag(S12_circle), 'c', 'LineWidth', 2);  
plot(real(nS21_circle), imag(nS21_circle), 'r', 'LineWidth', 2);
```

```
plot(real(C1), imag(C1), 'c+', 'MarkerSize', 10, 'LineWidth', 2);  
plot(real(C2), imag(C2), 'r+', 'MarkerSize', 10, 'LineWidth', 2);
```

```
%S12
```

```
str_S12 = {  
    '$S_{12} = C_1 - B e^{j(\theta_{12})}$';  
    sprintf('$ = (%.4f + j%.4f) - %.2f  
e^{j(%d^{\circ} + \theta_{12})}$', real(C1), imag(C1), B_val,  
theta_z_deg);  
    sprintf('$ = %.2f \angle %d^{\circ} -  
%.2f \angle (%d^{\circ} + \theta_{12})$', C1_mag, theta_z_deg, B_val,  
theta_z_deg);  
    ['$ = \left\{ \begin{array}{l} ' ...  
    sprintf('%.2f \angle %.2f^{\circ} \quad ; \quad (\theta_{12} = 0^{\circ})',  
abs(S12_zero), rad2deg(angle(S12_zero))) ...  
    ' \quad ' ...  
    sprintf('%.4f \angle %.2f^{\circ} \quad ; \quad (\theta_{12} = %d^{\circ})',  
abs(S12_max_point), rad2deg(angle(S12_max_point)),  
round(theta_P12max)) ...  
    ' \end{array} \right. $']  
};
```

```
text(2.5, 4.0, str_S12, 'Interpreter', 'latex', 'FontSize', 7, ...
```

```

    'BackgroundColor', 'w', 'EdgeColor', 'none', 'Margin', 3,
    'VerticalAlignment', 'top');

text(real(C1), imag(C1)+0.7, ...
    {sprintf('C_1 = %.2f \angle %d^\{\circ\}', C1_mag, theta_z_deg)},
    'Interpreter', 'tex', 'FontSize', 9);

text(real(C1), imag(C1)+0.5, sprintf('B=%.2f \angle %d^\{\circ\}',
    B_val, theta_z_deg), 'Interpreter', 'tex', 'FontSize', 9);

% P12 max
plot(real(S12_max_point), imag(S12_max_point), 'b.', 'MarkerSize', 15);
text(real(S12_max_point)+0.2, imag(S12_max_point), ...

sprintf('S_{12}=%.2f\angle%.2f^\{\circ\}\nP_{12max}=%.4f\n(\theta_{12}
=%d^\{\circ\})', ...
    abs(S12_max_point), rad2deg(angle(S12_max_point)), P12_max_val,
    round(theta_P12max)), ...
    'Interpreter', 'tex', 'FontSize', 9);

% theta12=0
plot(real(S12_zero), imag(S12_zero), 'b.', 'MarkerSize', 15,
    'MarkerFaceColor', 'w');
text(real(S12_zero)+0.15, imag(S12_zero), ...
    sprintf('%.2f\angle%.2f^\{\circ\}\n(\theta_{12}=0^\{\circ\})', ...
    abs(S12_zero), rad2deg(angle(S12_zero))), 'Interpreter', 'tex',
    'FontSize', 9);

% -S21
str_S21 = {
    '$-S_{21} = C_2 + B e^{j(- \theta_{12})}$';

```

```

    sprintf('$ = -(%.4f + j%.4f) + %.2f e^{j(%d^{\circ}-\theta_{12})}$', -real(C2), -imag(C2), B_val, theta_z_deg);
    sprintf('$ = -%.2f\angle%d^{\circ} + %.2f\angle(%d^{\circ}-\theta_{12})$', abs(C2), theta_z_deg, B_val, theta_z_deg);
    ['$ = \left\{ \begin{array}{l} ' ...
        sprintf('%.2f\angle%.2f^{\circ} \; ; \; (\theta_{12}=0^{\circ})',
abs(nS21_zero), rad2deg(angle(nS21_zero))) ...
        ' \; ' ...
        sprintf('%.4f\angle%.2f^{\circ} \; ; \; (\theta_{12}=%d^{\circ})',
abs(nS21_max_point), rad2deg(angle(nS21_max_point)),
round(theta_nP21max)) ...
        ' \end{array} \right. $']
};

text(-4.5, -2.5, str_S21, 'Interpreter', 'latex', 'FontSize', 7, ...
    'BackgroundColor', 'w', 'EdgeColor', 'none', 'Margin', 3);

text(real(C2), imag(C2)-0.5, ...
    {sprintf('C_2 = -%.2f \angle %d^{\circ}', abs(C2), theta_z_deg)},
    'Interpreter', 'tex', 'FontSize', 9);

text(real(C2), imag(C2)-0.7, sprintf('B=%.2f \angle %d^{\circ}',
B_val, theta_z_deg), 'Interpreter', 'tex', 'FontSize', 9);

% -P21 max
plot(real(nS21_max_point), imag(nS21_max_point), 'b.', 'MarkerSize',
15);
text(real(nS21_max_point)+0.2, imag(nS21_max_point), ...
    sprintf('-S_{21}=%.2f\angle%.2f^{\circ}\n-
P_{21max}=%.4f\n(\theta_{12}=%d^{\circ})', ...
    abs(nS21_max_point), rad2deg(angle(nS21_max_point)), nP21_max_val,
round(theta_nP21max)), ...
    'Interpreter', 'tex', 'FontSize', 9);

% 標記 theta=0

```

```
plot(real(nS21_zero), imag(nS21_zero), 'b.', 'MarkerSize', 15,  
      'MarkerFaceColor', 'w');  
text(real(nS21_zero)-1.2, imag(nS21_zero), ...  
      sprintf('%.2f\\angle%.2f^{\\circ}\\n(\\theta_{12}=\\theta^{\\circ})', ...  
      abs(nS21_zero), rad2deg(angle(nS21_zero))), 'Interpreter', 'tex',  
      'FontSize', 9);  
  
hold off;
```